

SQL Query Explanation

Understanding BETWEEN with text: FIRST_NAME BETWEEN 'A' AND 'C'

Main lesson

This query compares the whole text value, not only the first letter. That is why names like Adam and Bob can appear, but Charles usually does not appear when the upper limit is exactly 'C'.

```
SELECT *  
FROM employees  
WHERE FIRST_NAME BETWEEN 'A' AND 'C';
```

1. The Query

The query asks SQL to return all columns from the employees table, but only for rows where FIRST_NAME is between the text value A and the text value C.

```
SELECT *  
FROM employees  
WHERE FIRST_NAME BETWEEN 'A' AND 'C';
```

The condition is exactly the same as writing:

```
WHERE FIRST_NAME >= 'A'  
AND FIRST_NAME <= 'C';
```

Important rule

BETWEEN is inclusive. This means it includes the lower value and the upper value. So 'A' and 'C' themselves are included if they exist in the table.

2. What Does SQL Compare?

SQL does not look only at the first letter. It compares the whole text value alphabetically, character by character.

For example, SQL compares Adam with A and C. It also compares Charles with A and C. The full word matters.

Name	Is it >= 'A'?	Is it <= 'C'?	Result in original query
A	YES	YES	Included
Adam	YES	YES	Included
Alice	YES	YES	Included
Bob	YES	YES	Included
C	YES	YES	Included
Charles	YES	NO	Not included
Chris	YES	NO	Not included
David	YES	NO	Not included

Why Charles is not included:

The upper limit is exactly 'C'. The word 'Charles' starts with C, but it has more letters after C. In alphabetical comparison, 'C' comes before 'Charles'. So:

```
'Charles' > 'C'  
Therefore: 'Charles' <= 'C' is FALSE
```

3. Simple Dictionary Example

Imagine the names are ordered like a dictionary:

Dictionary order
A
Adam
Alice
B
Bob
C
Charles
Chris
D
David

When the condition says `<= 'C'`, SQL stops at the value C itself. It does not continue to Charles or Chris.

Beginner mistake

`BETWEEN 'A' AND 'C'` does not mean: names starting with A, B, or C. It means: the complete name must be alphabetically between the full text value 'A' and the full text value 'C'.

4. Better Query for Names Starting with A, B, or C

If your real goal is to get names that start with A, B, or C, use a range that starts at A and stops before D.

```
SELECT *  
FROM employees  
WHERE first_name >= 'A'  
AND first_name < 'D';
```

This includes names starting with A, B, and C, because all of them come before D.

Name	Condition: >= 'A' and < 'D'	Reason
Adam	Included	Starts with A
Alice	Included	Starts with A
Bob	Included	Starts with B
Charles	Included	Starts with C
Chris	Included	Starts with C
David	Not included	Starts with D, so it is not before D
Steven	Not included	Starts after D

5. Another Clear Way: SUBSTR

You can check only the first character by using SUBSTR. This is very clear for beginners because it directly says: take the first letter.

```
SELECT *
FROM employees
WHERE SUBSTR(first_name, 1, 1) BETWEEN 'A' AND 'C';
```

Meaning:

- SUBSTR(first_name, 1, 1) takes the first letter of the name.
- BETWEEN 'A' AND 'C' checks if that first letter is A, B, or C.

first_name	First letter	Included?
Adam	A	YES
Bob	B	YES
Charles	C	YES
David	D	NO

Performance note

SUBSTR is easy to understand, but using a function on a column can make normal indexes less useful in some databases. For learning, it is fine. For production systems, the range query can be better.

6. Another Easy Way: LIKE

LIKE is simple when you want names that start with specific letters.

```
SELECT *
FROM employees
WHERE first_name LIKE 'A%'
OR first_name LIKE 'B%'
OR first_name LIKE 'C%';
```

The percent sign % means: any characters after this.

Pattern	Matches examples
'A%'	Adam, Alice, Ahmed
'B%'	Bob, Brandon
'C%'	Charles, Chris

7. Which Option Should You Use?

Goal	Recommended query	Why
Full text is between A and C	<code>first_name BETWEEN 'A' AND 'C'</code>	Compares the complete text value.
Names starting with A, B, or C	<code>first_name >= 'A' AND first_name < 'D'</code>	Clean range condition and usually efficient.
Very clear beginner logic	<code>SUBSTR(first_name, 1, 1) BETWEEN 'A' AND 'C'</code>	Checks only the first letter directly.
Simple readable filters	<code>LIKE 'A%' OR LIKE 'B%' OR LIKE 'C%'</code>	Easy to understand, especially for small examples.

Best answer for this lesson

For names starting with A, B, or C, the cleanest query is usually: `WHERE first_name >= 'A' AND first_name < 'D'`.

8. Oracle and MySQL Notes

- In Oracle, `SUBSTR(column, start, length)` is commonly used.
- In MySQL, `SUBSTR()` also works, and `SUBSTRING()` is another common name for the same idea.
- Text comparison can be affected by database collation and case-sensitivity settings. For beginner examples, focus first on the logic: full text comparison vs first-letter comparison.

9. Practice Examples

Try to answer before checking the solution.

Input name	<code>BETWEEN 'A' AND 'C'</code>	<code>>= 'A' AND < 'D'</code>	Why
A	Included	Included	The exact lower value is included.
Amr	Included	Included	Starts with A, before C and before D.
C	Included	Included	The exact upper value C is included in BETWEEN.
Carlos	Not included	Included	Carlos is greater than 'C', but before 'D'.
Dina	Not included	Not included	Starts with D.

10. Final Summary

- BETWEEN 'A' AND 'C' is the same as `>= 'A' AND <= 'C'`.
- BETWEEN is inclusive, so it includes the exact values A and C.
- When used with text, SQL compares the full value, not only the first letter.
- Charles is not included in BETWEEN 'A' AND 'C' because Charles is greater than C.
- For names starting with A, B, or C, use `first_name >= 'A' AND first_name < 'D'`.

One-line memory rule

BETWEEN with text compares the whole word. To include all names starting with C, stop before D, not at C.

Clean final query:

```
SELECT *  
FROM employees  
WHERE first_name >= 'A'  
AND first_name < 'D';
```